Unit 2 FRQ Set 3

Forces & Translational Dynamics · 4 free-response questions in real AP exam order

40 points Time: **95 min** (the AP exam's FRQ section) MR 10 · TBR 12 · LAB 10 · QQT 8

- Read all four questions first (2 min). Start with the one you're most sure of.
- **Derive** = start from a law in symbols. **Calculate** = equation, numbers in, units out. **Justify** = claim → law → link to THIS situation.
- Never leave a part blank. Points are awarded line by line.
- When the timer ends, stop. Grade yourself with the coaching guide's rubric, and be strict.

Reference (same ideas as the AP equation sheet)

$\Sigma F = ma$ · $F_g = mg$ · $F_g = Gm_1m_2/r^2$ · $g = GM/r^2$ · $v = v_0 + at$

$f_s \le \mu_s F_N$ · $f_k = \mu_k F_N$ · $a_c = v^2/r$ · $v = 2\pi r/T$ · $\Delta x = v_0 t + \frac{1}{2}at^2$

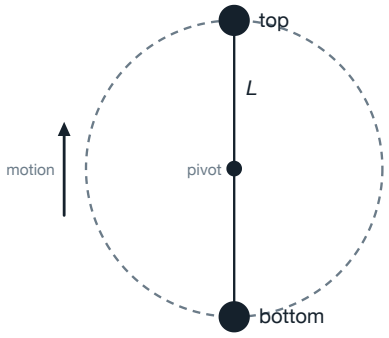
$x_{cm} = \Sigma m_i x_i / \Sigma m_i$ · $F_s = -k\Delta x$ · $v^2 = v_0^2 + 2a\Delta x$

$g = 9.8\ m/s^2$ · $G = 6.67 \times 10^{-11}\ N \cdot m^2/kg^2$

Question 1: Mathematical Routines (MR)

10 points · suggested 20–25 min

A small ball of mass m on a string of length L is swung in a vertical circle. At the bottom of the circle the ball has speed v_b; at the top it has speed v_t.



A. On the two dots below, draw and label the forces exerted on the ball (i) at the top of the circle and (ii) at the bottom of the circle. (2 pts)

(i) top

(ii) bottom

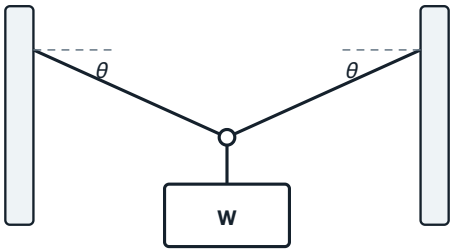
B. Derive an expression for the tension in the string when the ball is at the bottom of the circle, in terms of m, v_b, L and physical constants, as appropriate. (3 pts)

C. For m = 0.20 kg, L = 0.80 m and v_b = 4.0 m/s, calculate (i) the tension at the bottom and (ii) the minimum speed the ball can have at the top and still move in a circle. `(2 pts)`

D. The string breaks if its tension exceeds 8.0 N. Where in the circle is the string most likely to break? Justify your answer, and calculate the maximum speed the ball can have at that point. `(3 pts)`

Question 2: Translation Between Representations (TBR) 12 points · suggested 25–30 min

A sign of weight W hangs at rest from a light ring. Two identical cables hold the ring, each making angle θ with the horizontal, as shown.



A. On the dot below, which represents the ring, draw and label the forces exerted on the ring. Draw the forces at the correct angles. (3 pts)

B. Derive an expression for the tension T in each cable in terms of W and θ. (3 pts)

C. On the axes below, sketch T as a function of θ for angles from about 10° to 90°. Label the value of T at θ = 90°. (4 pts)

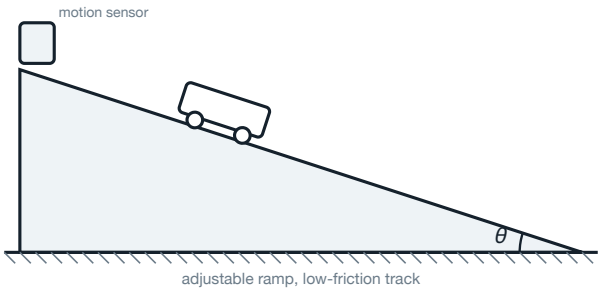


D. A student claims: “If you pull the cables hard enough, you can make them perfectly horizontal while the sign is hanging from them.” Use your answers to parts B or C to explain why the student is incorrect. `(2 pts)`

Question 3: Experimental Design and Analysis (LAB)

10 points · suggested 25–30 min

Students want to determine the gravitational field strength g using a low-friction cart on an adjustable ramp. They have a meterstick, a protractor, a stopwatch and a motion sensor.

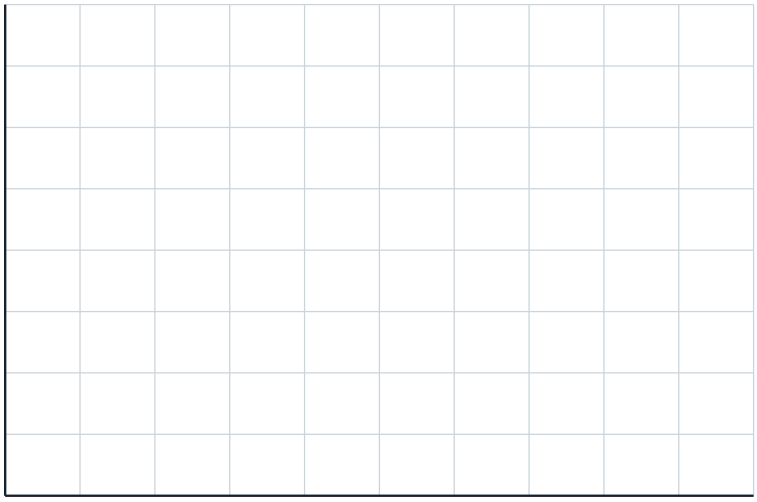


A. Describe a procedure to collect data that could be graphed to determine g. Include what is varied, what is measured, and how. (2 pts)

B. The data are shown below. Indicate what quantities should be graphed to produce a straight line whose slope is g. (2 pts)

Ramp angle θ (°)	sin θ	Acceleration a (m/s²)
5		0.86
10		1.69
15		2.55
20		3.33
25		4.16

C. On the grid below, plot the data, label the axes with units, choose an appropriate scale and draw a best-fit line. `(4 pts)`

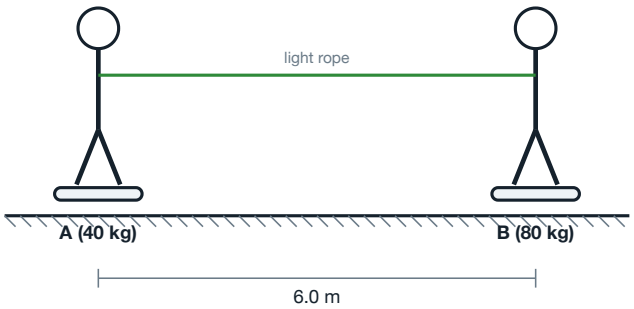


D. Use the best-fit line to calculate g, and give one reason your value might be slightly less than the accepted 9.8 m/s^2. `(2 pts)`

Question 4: Qualitative/Quantitative Translation (QQT)

8 points · suggested 15–20 min

Skater A (40 kg) and Skater B (80 kg) stand at rest on frictionless ice, 6.0 m apart, holding the ends of a light rope. Skater B pulls the rope hand over hand until the skaters meet.



A. WITHOUT using equations, explain where the skaters meet: closer to A's starting position, closer to B's, or exactly in the middle. (3 pts)

B. Calculate how far from A's starting position the skaters meet. (3 pts)

C. Explain how your calculation supports your reasoning in part A. Would the answer change if Skater A did the pulling instead? (2 pts)